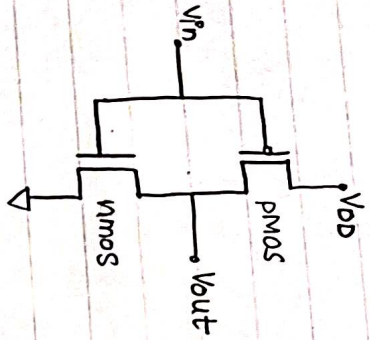


PRACTICAL NO:-02

AIM:- Implementation of CMOS Inverter.

SOFTWARE:- LTSPICE

DIAGRAM:-



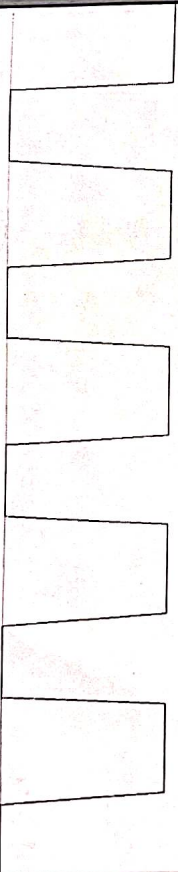
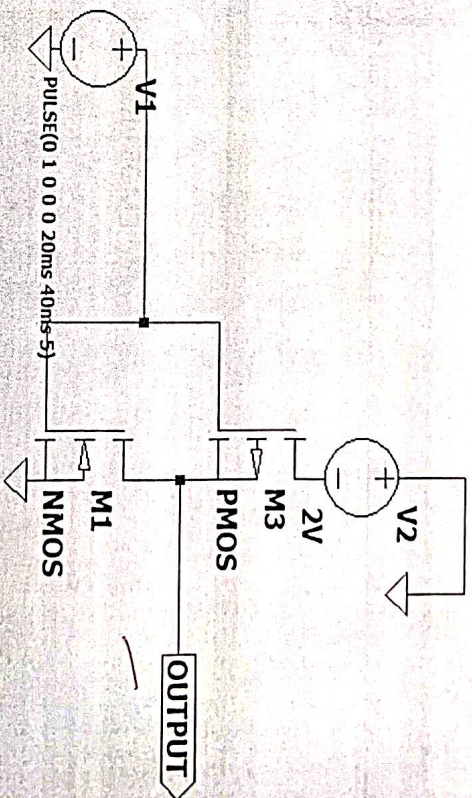
TRUTH TABLE:-

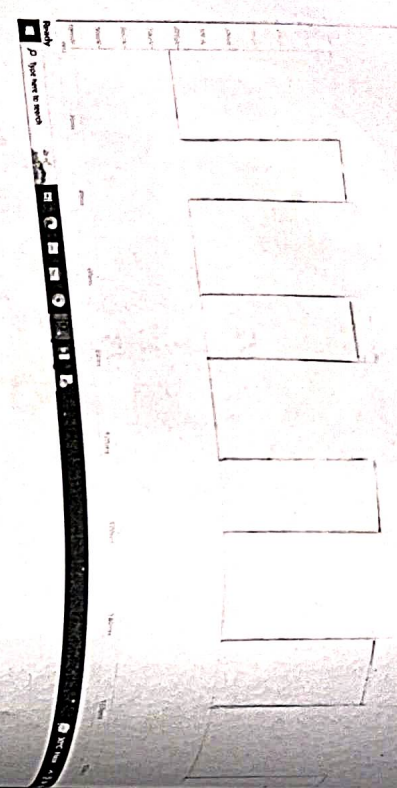
V_{in}	V_{out}
0	1
1	0

PRACTICAL NO.:-02

AIM:- Implementation of Inverter using CMOS.

SOFTWARE:- LTSPICE





Result:- Thus, We have successfully implemented CMOS inverter.

PRACTICAL NO:-02

● **AIM:-** Implementation of CMOS inverter.

● **SOFTWARE:-**

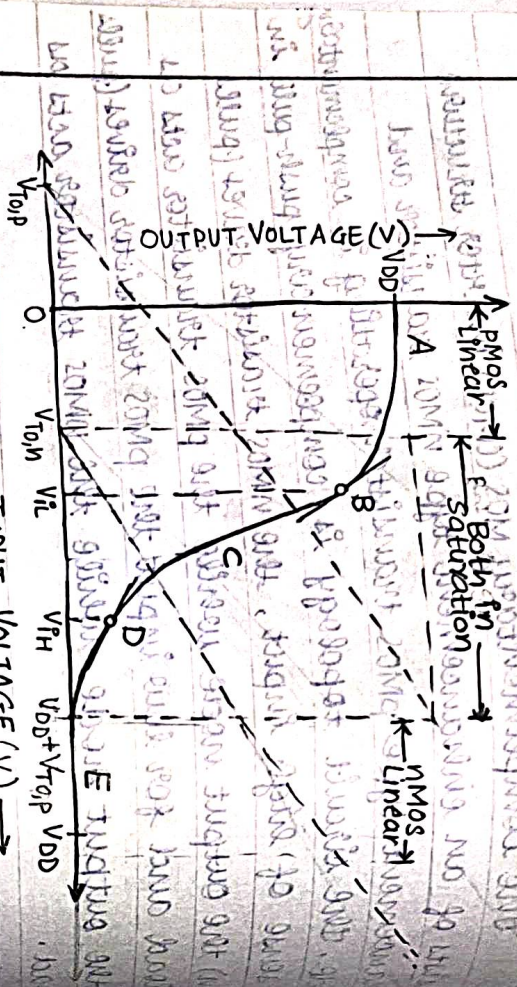
● **THEORY:-**

Inverter is a logical operator which produces a output exact opposite of input. That is, when the input is high the output becomes low and when the input is low the output becomes high.

The complementary MOS (CMOS) inverter structure consists of an enhancement type NMOS transistor and enhancement type PMOS transistor operating in complementary mode. The circuit topology is complementarily push-pull in the sense of high input, the NMOS transistor deliver (pulls down) the output node while the PMOS transistor acts as a load and for low input the PMOS transistor delivers (pulls up) the output node while the NMOS transistor acts as a load.

The CMOS inverter has the important advantages over other inverter configurations that are:-

- (1) the steady state power dissipation of the CMOS inverter is virtually negligible, except for small power dissipation due to leakage currents.
- (2) the voltage transfer characteristics (VTC) exhibits a full output voltage swing between 0V and V_{DD} and that the VTC transition is usually very sharp. Thus, the VTC of the CMOS inverter resembles that of an ideal inverter.



VI characteristics of CMOS Inverter

- Circuit Operation :-** The input voltage is connected to the gate terminal of both the NMOS and the PMOS transistors. Thus, both transistors are driven directly by the input signal V_{in} . The substrate of the NMOS transistor is connected to ground and the substrate of the PMOS transistor is connected to power supply voltage, V_{DD} , in order to reverse bias the source and drain junctions. Since $V_{GS} = 0$ for both devices, there will be no substrate bias effect for either device.
- The CMOS inverter works in five different regions and this gives us a full voltage swing between 0V and V_{DD} . The five regions since the VI characteristics of inverter and its operation in five different regions. These regions are :-
- Region A :-** When the inverter is in region A, the PMOS transistor functions in the linear region and NMOS transistor functions in the activation region because $V_{in} > V_{DD}$ and V_{th} and the output equals to V_{DD} ($V_{OH} = V_{DD}$).
 - Region B :-** The NMOS transistor starts conducting in activation mode and the output voltage decreases. Also, the critical voltage V_i corresponds to $(dV_{out}/dV_{in}) = -1$.
 - Region C :-** In region C, the voltage further decreases and the PMOS transistor enters activation region. At this point both the transistors operate in activation region as the voltage has attained threshold component.

- (4) **Region D:-** When the transistor enters in region D, the MOS transistor starts working in linear mode. Because the input voltage V_{in} becomes less than the threshold voltage. At this point the critical voltage V_{th} corresponds to $(dV_{out}/dV_{in}) = -1$.
- (5) **Region E:-** In region E, the MOS transistor operates in linear mode and the MOS transistor operates in cutoff region because the input voltage $V_{in} > (V_{DD} + V_{th})$. Here, the output equals to 0 ($V_{out} = 0$).

● **RESULT:-** Thus, we successfully implemented CMOS inverters.

D	K	Total
01	01	01
02	02	02
03	03	03
04	04	04
05	05	05
06	06	06
07	07	07
08	08	08
09	09	09
10	10	10
11	11	11
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94	94	94
95	95	95
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98	98	98
99	99	99
100	100	100

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